Elena Zoghby

Telecommunications & Computer Engineering Student | AI, Machine Learning & TinyML

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EDUCATION

- **Lebanese University, Faculty of Engineering – Branch 1, Ras Maska, LB** 2022–present
  Fifth-Year Telecommunications & Computer Engineering student (Expected Graduation: 2027)
- **Bishmizzine High School, Bishmizzine, LB** 2022
  Lebanese Baccalaureate in General Sciences

EXPERIENCE

AI Fellowship Program | *LebNet* | *Remote, LB* 06/2026–08/2026

- Graduated with High Distinction; covered ML foundations, transformers, LLM fine-tuning, agents, and RAG
- Capstone: reproduced RepViT (CVPR 2024) and built an early-exit inference mechanism that uses the confidence (softmax output) of the penultimate stage to skip the final stage
- Reached 80.34% accuracy, within 0.06 points of the full model, while skipping the final stage on ~17% of inputs

Full Stack Development Intern (Team Lead) | *TechTalks* | *Remote, LB* 02/2026–03/2026

- Led development of a Smart Expense Tracker web application
- Built backend logic, UI components, and database integration

KEY PROJECTS

AI-Based Hand Gesture Recognition and Audio Output System | *TinyML, TensorFlow Lite Micro, ESP32-CAM, Python*

- Built an end-to-end TinyML system: a custom CNN quantized to INT8 and deployed on ESP32-CAM, recognizing 6 hand gestures plus a "no gesture" class (~90.8% validation accuracy)
- Captured 7,000+ images across three collection phases and curated a final 2,000-image dataset; compared multiple CNN architectures against hardware limits
- Python application receives predictions over serial and plays audio feedback; supervised by Prof. Wassim Falou and Prof. Mayssa Hajar

Sign Safe | *Lovable, Groq API*

- Solo-built AI web app that reads a job offer, flags risky contract terms by severity (low/medium/high), and benchmarks compensation against industry norms
- Built for the Kanz AI Hackathon (LAU, July 2026); awarded a Gold rating

Link: https://sign-safebuddy.lovable.app/

E-Commerce Desktop Application | *C#, SQL, Visual Studio*

- Engineered a desktop application with product inventory management and a relational database backend

ACHIEVEMENTS & HACKATHONS

Kanz AI Hackathon, Lebanese American University (LAU) July 2026

- Part of the Guinness World Record for the largest AI video lesson (14,075 participants)
- Built "Sign Safe" (see Key Projects), which received a Gold rating from hackathon judges

IEEEXtreme 24-Hour Programming Competition

- Solved complex algorithmic problems under time constraints using Python

TECHNICAL SKILLS

- **AI & ML:** CNNs, computer vision, transformers, LLM fine-tuning, RAG, AI agents, LLM API integration (Groq), model optimization (INT8 quantization, early-exit inference)
- **TinyML & Edge AI:** TensorFlow Lite Micro, ESP32-CAM, on-device inference
- **Programming Languages:** Python, Java, C#, SQL
- **Tools & Platforms:** Git/GitHub, VS Code, Visual Studio, Jira, Postman, Notion, Groq API, Lovable